

List 1 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

ANS ::

1. **No Display**

Symptom: Blank screen or no display on the monitor.

2. **RAM Problem**

Symptom: Computer does not start or produces beep sounds.

3. **Hard Disk/SSD Failure**

Symptom: Computer becomes slow or does not boot.

4. **Overheating**

Symptom: Computer becomes very hot or shuts down suddenly.

5. **Power Supply Problem**

Symptom: Computer does not turn on or restarts suddenly.

2 Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

1. Check the Power Socket

Check whether the wall socket is working by plugging in another device.

2. Check the Charger

Inspect the charger, cable, and adapter for any cuts, damage, or loose connections.

3. Check the Charging Port

Look inside the laptop charging port for dust, dirt, or physical damage.

4. Reconnect the Charger

Disconnect the charger and plug it back in firmly. Check whether the charging light turns on.

5. Check the Battery

If possible, remove and reconnect the battery. Look for swelling or physical damage.

6. Test with Another Compatible Charger

Use a known-working compatible charger to check whether the original charger is faulty.

7. Check the Result

If the laptop still does not charge, the problem may be with the **battery, charging port, or internal power circuit** and it should be checked by a technician.

3 Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

MULTIMETER TEST USB PHONE CHARGER



➤ Multimeter Test – USB Phone Charger

1. Set the multimeter to **DC Voltage (V=)**.
2. Connect the **black probe** to the negative (-) contact.
3. Connect the **red probe** to the positive (+5V) contact.
4. Check and record the voltage shown on the multimeter.

Measured Voltage: 5.0 V

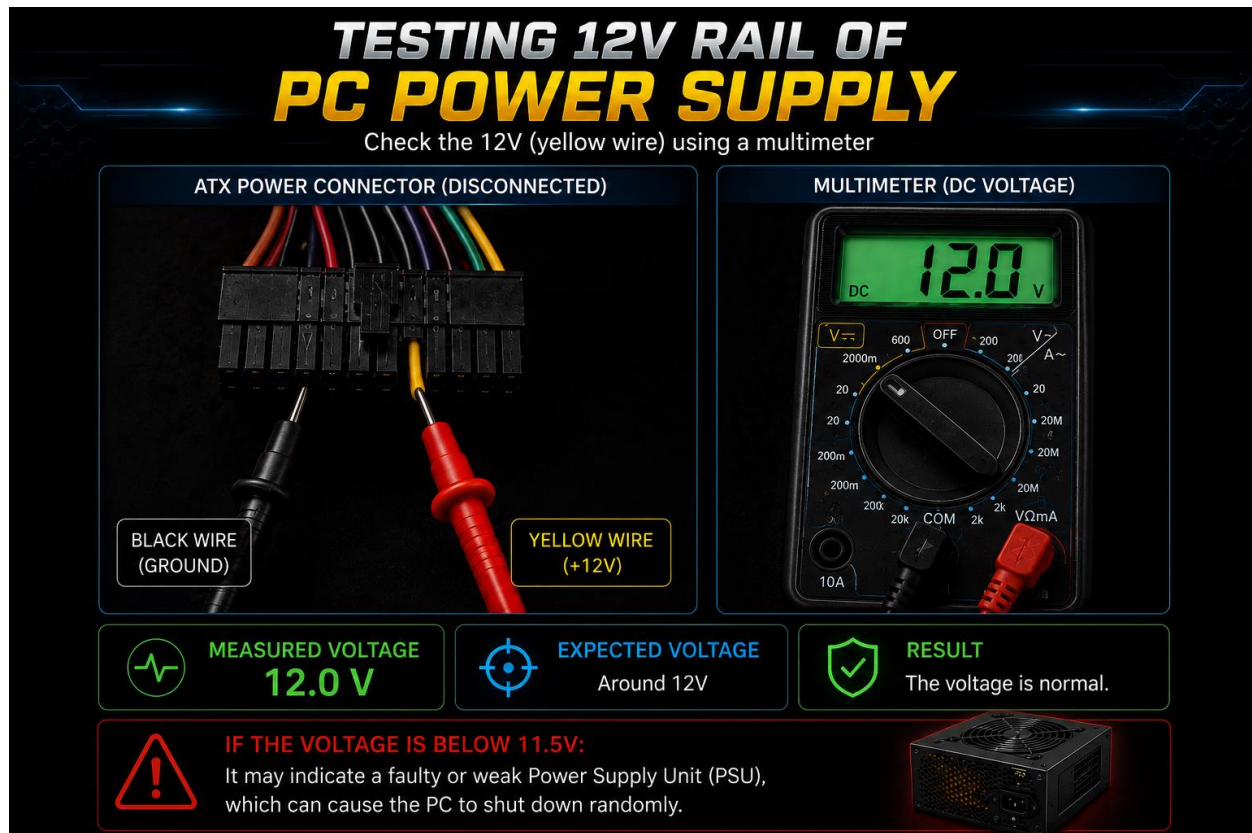
Expected Range: 4.75 V – 5.25 V

Result: The voltage is within the expected range, so the **charger is working normally**.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this

indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.



Power Supply 12V Rail Test

1. **Turn off the PC** and unplug it from the power source.
2. Disconnect the **ATX power connector** from the motherboard.
3. Set the multimeter to **DC Voltage (V=)**.
4. Connect the **black probe** to a **black wire (Ground)**.
5. Connect the **red probe** to the **yellow wire (+12V)**.

6. Check and record the voltage shown on the multimeter.

Measured Voltage: 12.0 V

Expected Voltage: Around 12 V

Result: The voltage is normal.

If the voltage is below 11.5 V: It may indicate a **faulty or weak Power Supply Unit (PSU)**, which could cause the PC to shut down randomly.